Taxonomists have assigned a method of classifying organisms which are called **ranks.** These taxonomical ranks are hierarchical. The scheme of classification has to be flexible, allowing newly discovered living organisms to be added where they fit best.

5.9.1 Need for Classification

- Understanding the classification of organisms can gives an insight in to other fields and has significant practical value.
- Classification helps us to know about different taxa, their phylogenetic relationship and exact position.
- It helps to train the students of plant sciences with regard to the diversity of organisms and their relationship with other biological branches.

5.10 Types of classification

Taxonomic entities are classified in three ways. They are artificial classification, natural classification and phylogenetic classification.

5.10.1 Artificial system of classification

Carolus Linnaeus (1707 -1778) was a great Swedish Botanist and said to be the "**Father of Taxonomy**." He outlined an artificial system of classification in "***Species Plantarum***" in 1753, wherein he listed and described 7,300 species and arranged in 24 classes mostly on the basis of number, union



Figure 5.4: Carolus Linnaeus

24 classes recognized by Linnaeus in his *Species Plantarum* (1753) on the basis of stamens.

No	Classes	Characters
1	Monandria	stamen one
2	Diandria	stamens two
3	Triandria	stamens three
4	Tetrandria	stamens four
5	Pentandria	stamens five
6	Hexandria	stamens six
7	Heptandria	stamens seven
8	Octandria	stamens eight
9	Ennandria	stamens nine
10	Decandria	stamens ten
11	Dodecandria	stamens 11-19
12	Icosandria	stamens 20 or more, on the calyx
13	Polyandria	stamens 20 or more, on the receptacle
14	Didynamia	stamens didynamous; 2 short, 2 long
15	Tetradynamia	stamens tetradyna-mous; 4 long, 2 short
16	Monadelphia	stamens monadelphous; united in 1 group
17	Diadelphia	stamens diadelphous; united in 2 groups
18	Polyadelphia	stamens polyadelphous; united in 3 or more groups
19	Syngenesia	stamens syngenesious; united by anthers only
20	Gynandria	stamens united with the gynoecium
21	Monoecia	plants monoecious
22	Dioecia	plants dioecious
23	Polygamia	plants polygamous
24	Cryptogamia	flowerless plants